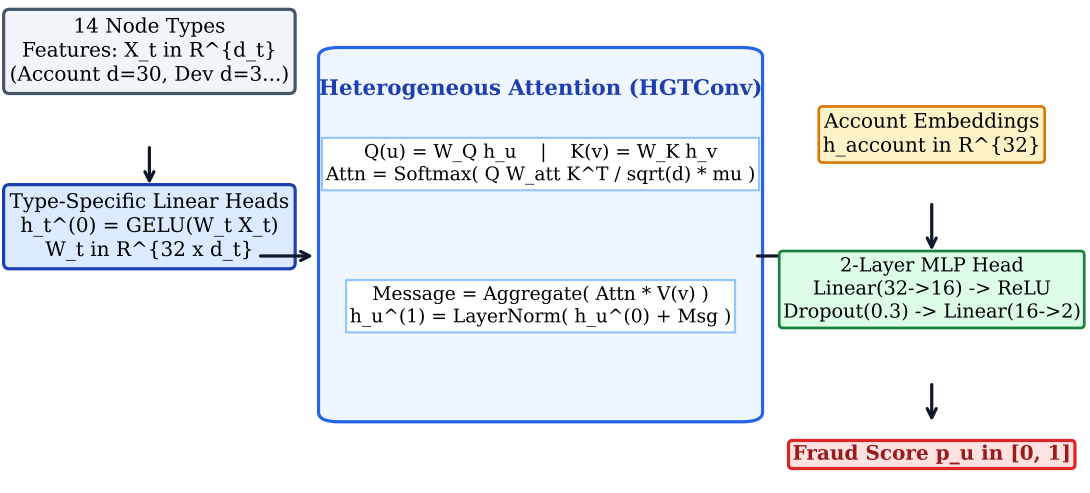
14 Node Types
Features: X_t in R^{d_t}
(Account d=30, Dev d=3...)
Type-Specific Linear Heads
h_t^(0) = GELU(W_t X_t)
W_t in R^{32 x d_t}
Heterogeneous Attention (HGTConv)
Q(u) = W_Q h_u | K(v) = W_K h_v
Attn = Softmax(Q W_att K^T / sqrt(d) * mu)
Message = Aggregate(Attn * V(v))
h_u^(1) = LayerNorm(h_u^(0) + Msg)
Account Embeddings
h_account in R^{32}
2-Layer MLP Head
Linear(32->16) -> ReLU
Dropout(0.3) -> Linear(16->2)
Fraud Score p_u in [0, 1]